

Predictive Biomarkers in Heart Failure: A Logistic Regression Approach

Master of Science Thesis Defense

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Methodological Framework

- **Objective:** Evaluate predictors of mortality using clinical records.
- **Model:** Binary Logistic Regression with stratified train-test split.
- **Equation:**

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_{EF} + \beta_2 X_{Cr} + \epsilon \quad (1)$$

- **Significance:** Model achieved an ROC-AUC of 0.862.

Key Findings

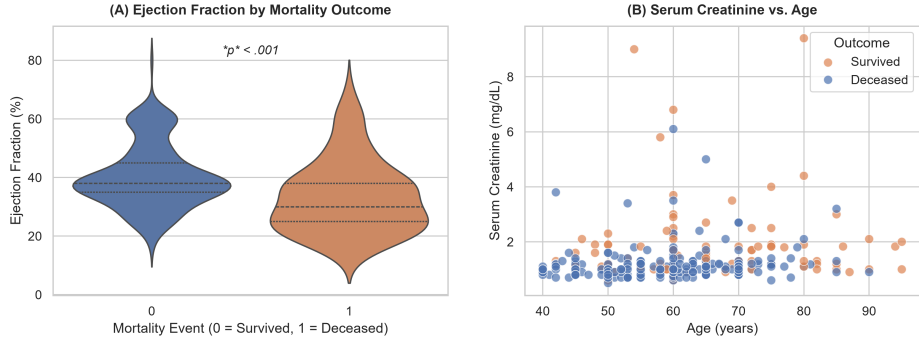


Figure: Distribution of Ejection Fraction and Serum Creatinine by outcome.

References I